

Assignment 3.

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1. Father, mother, 2 boys, 3 girls

(a) there are no restrictions.

$$7! = 5040 \text{ ways}$$

(b) parents stand together

FM B₁ B₂ G₁ G₂ G₃

$$= {}^6P_1 \times {}^5P_5 \times {}^2P_2$$

$$= 1440 \text{ ways}$$

(c) all female stand together

MG₁G₂G₃ F B₁ B₂

$$= 4! \times 3!$$

$$= 144 \text{ ways}$$

2. TUESDAY

(a) 4 letters are used at a time

$$= {}^7P_4$$

$$= 840 \text{ ways}$$

(b) all letters be used at a time

$$= {}^7P_7$$

$$= 5040 \text{ ways}$$

(c) all letters are used but first letter vowel

$$\text{Vowel} = 3$$

$$6! = 720$$

$$= 3 \times 720$$

$$= 2160 \text{ ways}$$

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3. registration number = 3 letters, four single digit numbers.

(a) letter cannot be repeated.

26 letter	10 digit
$26 \cdot 25 \cdot 24$	$10 \cdot 10 \cdot 10 \cdot 10$
$= 26 \times 25 \times 24$	$= 10 \times 10 \times 10 \times 10$
$= 15,600$	$= 10,000$

$$\text{Total registration} = 15,600 \times 10,000$$

$$= 156,000,000$$

(b) neither letter nor number can be repeated.

26 letter	10 digit
$26 \cdot 25 \cdot 24$	$= 10 \cdot 9 \cdot 8 \cdot 7$
$= 26 \times 25 \times 24$	$= 10 \times 9 \times 8 \times 7$
$= 15,600$	$= 5,040$

$$\text{Total registration} = 15,600 \times 5,040$$

$$= 78,624,000$$

4. committee of 7 = 9 boys & 4 girls.

(a) exactly 3 girls.

$$= {}^9C_4 \times {}^4C_3$$

$$= 504 \text{ ways}$$

(b) At least 3 girls.

$$({}^9C_4 \times {}^4C_3) + ({}^9C_3 \times {}^4C_4)$$

$$= 504 + 84$$

$$= 588 \text{ ways}$$

(c) at most 3 girls.

$$= ({}^9C_4 \times {}^4C_0) + ({}^9C_6 \times {}^4C_1) + ({}^9C_5 \times {}^4C_2) + ({}^9C_4 \times {}^4C_3)$$

$$= 36 + 336 + 756 + 504$$

$$= 1632 \text{ ways}$$

No.:

Date:

5

100 participants

3 prizes

grand prize

second prize

third prize

$${}^{100}C_3 = 161700$$

$${}^{99}C_2 = 4851$$

$$P(\text{Jumlah wins}) = \frac{4851}{161700}$$

$$= 0.03$$

(1) probability

$$(1) \frac{1}{c}$$

(2) probability

$$(2) \frac{1}{c}$$

total number of prizes = 1 + 1 + 1

$$= (1 + 1 + 1) \cdot 4$$

$$(1 + 1 + 1) \cdot 9 = (1 + 1 + 1) \cdot 1$$

$$2 = (1 + 1 + 1) \cdot 9 = 1 = (1 + 1 + 1)$$

$$3 = (1 + 1 + 1) \cdot 9 = 1 = (1 + 1 + 1)$$

$$1 = (1 + 1 + 1) \cdot 1 = (1 + 1 + 1)$$

2

3

total =

6.

1 bag

white	black
ball	ball
(4)	(3)

1 bag

white	black
ball	ball
(3)	(5)

randomly pick

probability pick ball is black from second bag.

Bag 1

white = 4

black = 3

total = 4 + 3

= 7

Bag 2

white = 3

black = 5

total = 3 + 5

= 8

probability (B_1)

$$= \frac{1}{2} (0.5)$$

probability (B_2)

$$= \frac{1}{2} (0.5)$$

total probability

Let D = event of picking black ball

 $P(B_2 \cap D) =$ picking ball from bag 2

$$P(B_2 \cap D) = P(B_2) \cdot P(D|B_2)$$

$$P(B_2) = \frac{1}{2}$$

$$P(D|B_2) = \frac{5}{8}$$

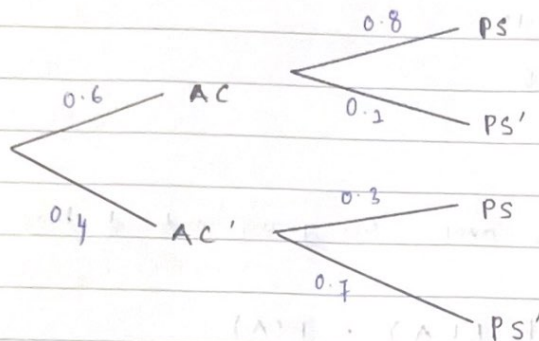
$$P(B_2) \cdot P(D|B_2) = \frac{1}{2} \cdot \frac{5}{8}$$

$$= \frac{5}{16}$$

$$= 0.3125$$

7. AC = advanced calculus

PS = pre-health science.



$$\begin{aligned}
 P(PS) &= (0.6 \times 0.8) + (0.4 \times 0.3) \\
 &= 0.48 + 0.12 \\
 &= 0.6
 \end{aligned}$$

$$P(PS | AC) = 0.8$$

$$P(PS | AC') = 0.3$$

$$P(AC | PS) = \frac{P(PS | AC)}{P(PS)}$$

$$\begin{aligned}
 &= \frac{(0.8)(0.6)}{0.6}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{0.48}{0.6}
 \end{aligned}$$

$$P(AC | PS) = 0.80$$

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8.

$$P(A) = 0.15$$

$$P(B|A) = 0.28$$

$$P(A') = 1 - P(A)$$

$$P(B|A') = 0.56$$

$$= 1 - 0.15$$

$$= 0.85$$

(a) the person is over 60 years old & has a loan

$$P(A \cap B) = P(B|A) \cdot P(A)$$

$$= 0.28 \cdot 0.15$$

$$= 0.042$$

(b) the person has no loan

$$P(B') = 1 - P(B)$$

$$P(B) = [P(B|A) \cdot P(A)] + [P(B|A') \cdot P(A')]$$

$$= (0.28 \cdot 0.15) + (0.56 \cdot 0.85)$$

$$= 0.042 + 0.476$$

$$= 0.518$$

$$P(B') = 1 - 0.518$$

$$= 0.482$$

(c) The person has loan, the person less than or equal to 60 years

$$P(B|A') = 0.56$$

(d) the person is over 60 & person has loan.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B)$$

$$= \frac{0.042}{0.518}$$

$$= 0.0811$$

9. 1. set of vertices, set of edges, table edge-endpoint.

$$V = \{ v_1, v_2, v_3, v_4, v_5, v_6 \}$$

$$e = \{ e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9, e_{10} \}$$

edge	end point
e_1	$\{v_1, v_2\}$
e_2	$\{v_2, v_3\}$
e_3	$\{v_3, v_4\}$
e_4	$\{v_4, v_5\}$
e_5	$\{v_4\}$
e_6	$\{v_4, v_5\}$
e_7	$\{v_5, v_2\}$
e_8	$\{v_1, v_5\}$
e_9	$\{v_1, v_5\}$
e_{10}	$\{v_1, v_6\}$

- ii- incidence matrix.

[illegible]

No.:

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10. $V_1 = e_4, e_6$

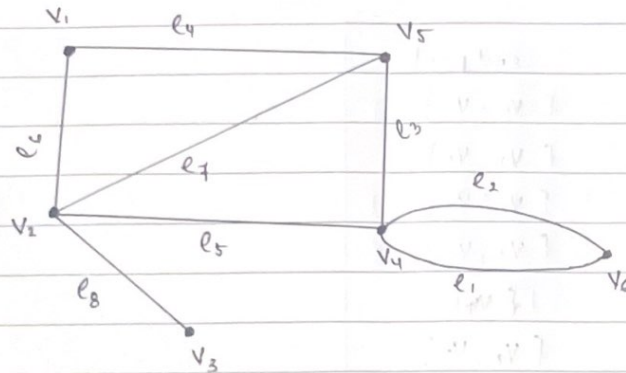
$V_4 = e_1, e_2, e_3, e_5$

$V_2 = e_5, e_6, e_7, e_8$

$V_5 = e_3, e_4, e_7$

$V_3 = e_8$

$V_6 = e_1, e_2$



11.

$$A_{G_2} = \begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 5 \\ \hline 1 & 0 & 0 & 0 & 1 & 0 \\ 2 & 0 & 0 & 1 & 0 & 1 \\ 3 & 0 & 1 & 0 & 1 & 1 \\ 4 & 1 & 0 & 1 & 0 & 1 \\ 5 & 0 & 1 & 1 & 1 & 0 \end{array}$$

$$A_{G_3} = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 3 & 0 & 1 & 0 & 0 \\ 4 & 0 & 1 & 1 & 0 \end{array}$$

14.

Graph 1

$$V = 4$$

$$W = 7$$

$$\deg(A) = 4$$

$$\deg(B) = 3$$

$$\deg(C) = 4$$

$$\deg(D) = 1$$

$$f(A) = A'$$

$$f(B) = B'$$

$$f(C) = C'$$

$$f(D) = D'$$

Graph 2

$$V = 4$$

$$W = 7$$

$$\deg(A') = 4$$

$$\deg(B') = 3$$

$$\deg(C') = 4$$

$$\deg(D') = 1$$

- Both Graph 1 and graph 2 has 4 vertices and 7 edges
- Both Graph 1 and graph 2 has same degree for corresponding vertices
- Both Graph 1 and graph 2 has same numbers of connected of components
- Both Graph 1 and graph 2 have loops and parallel edges
- Both Graph 1 and graph 2 are connected graph
- $f(A) = A'$, $f(B) = B'$, $f(C) = C'$ and $f(D) = D'$